Out of these considerations a paradoxical conclusion was derived that Achilles could never catch up with the turtle.

This "paradox" shows that ancient Greeks failed to grasp the fact that a monotonic sequence may be bounded.

Imp

Reader: One has to agree that the presence of both the monotonicity and boundedness is something not so simple to understand.

Author: Indeed, this is not so simple. It brings us close to a discussion on the limit of sequence. The point is that if a sequence is both monotonic and bounded, it should necessarily have a limit.

Actually, this point can be considered as the "beginning" of calculus.

Sam: Some resources on Zeno's Paradoxes for later reading

1. Physics — Aristotle

This is where Zeno's paradoxes are most clearly discussed in antiquity.

- Aristotle analyzes and critiques Zeno's arguments.
- Discusses infinity (potential vs actual)
- Connects motion, time and divisibility.

2. Zeno's Paradoxes — Wesley C. Salmon

This is one of the best modern books on the topic.

- Explains each paradox clearly.
- Covers historical responses.
- Bridges philosophy and mathematics.
- Discusses how calculus resolves (or doesn't fully resolve) them.

3. Infinity and the Mind — Rudy Rucker

- Explores infinity in mathematics and philosophy.
- Connects Zeno to : i) calculus ii) set theory iii) paradoxes of infinity
- Very imaginative and thought-provoking.